

Table 1. Five-replicate comparison of CLCRN-AGF and the schedule-matched CLCRN control. Values are mean $\pm$ sample standard deviation across seeds 2021–2025. Differences are CLCRN-AGF minus control; negative values favour CLCRN-AGF. Confidence intervals and two-sided p -values use Welch’s unequal-variance approximation.

Task	Metric	Control (mean $\pm$ s)	CLCRN-AGF (mean $\pm$ s)	Difference [95% CI]	p_{Welch}
Temperature	MAE	1.258 $\pm$ 0.060	1.291 $\pm$ 0.107	+0.033 [−0.100, +0.166]	0.5701
	RMSE	2.101 $\pm$ 0.045	2.118 $\pm$ 0.129	+0.017 [−0.141, +0.174]	0.7942
Relative humidity	MAE	4.648 $\pm$ 0.017	4.537 $\pm$ 0.039	−0.111 [−0.158, −0.063]	0.0015
	RMSE	7.248 $\pm$ 0.043	7.010 $\pm$ 0.055	−0.239 [−0.311, −0.166]	8.17×10^{-5}
Surface wind	MAE	1.242 $\pm$ 0.011	1.256 $\pm$ 0.037	+0.014 [−0.032, +0.059]	0.4699
	RMSE	1.972 $\pm$ 0.023	2.016 $\pm$ 0.093	+0.043 [−0.071, +0.158]	0.3657
Cloud cover	MAE	0.1479 $\pm$ 0.0017	0.1484 $\pm$ 0.0020	+0.0005 [−0.0022, +0.0031]	0.7077
	RMSE	0.2500 $\pm$ 0.0017	0.2490 $\pm$ 0.0028	−0.0010 [−0.0045, +0.0025]	0.5103

All scores include valid zero targets and use equal spatial weights over the full 12-hour forecast and 657 test examples. Errors are in K for temperature, percentage points for relative humidity, m s^{-1} for surface wind, and fractions for cloud cover. Benjamini–Hochberg correction across the eight comparisons retains only the two humidity differences at $q < 0.05$ (MAE $q = 0.00604$; RMSE $q = 0.000653$).

Table 2. Contextual test results on the same test split. Persistence is deterministic; control and CLCRN-AGF are means across five seeds. All methods use the same zero-inclusive, spatially unweighted evaluation over the full 12-hour forecast. Lower is better.

Method	Temperature		Relative humidity		Surface wind		Cloud cover	
	MAE	RMSE	MAE	RMSE	MAE	RMSE	MAE	RMSE
Persistence	1.421	2.700	6.425	10.447	1.537	2.535	0.1620	0.2725
CLCRN control (5 seeds)	1.258	2.101	4.648	7.248	1.242	1.972	0.1479	0.2500
CLCRN-AGF (5 seeds)	1.291	2.118	4.537	7.010	1.256	2.016	0.1484	0.2490

Errors are in K for temperature, percentage points for relative humidity, m s^{-1} for surface wind, and fractions for cloud cover. Variation across model seeds is reported in Table 1.

Table 3. Component ablations with a matched 100-epoch FP32 training protocol. All entries report mean $\pm$ sample standard deviation over seeds 2023–2025 on the full 12-hour test forecast. Lower is better.

Budget	Task	Variant	MAE	RMSE
100 epochs, 3 seeds	Temperature	Control (no AGF)	1.233 ± 0.056	2.113 ± 0.043
		w/o gated fusion	1.267 ± 0.066	2.047 ± 0.103
		Full CLCRN-AGF	1.250 ± 0.127	2.057 ± 0.138
100 epochs, 3 seeds	Relative humidity	Control (no AGF)	4.636 ± 0.009	7.227 ± 0.043
		w/o gated fusion	4.599 ± 0.024	7.132 ± 0.062
		Full CLCRN-AGF	4.555 ± 0.041	7.040 ± 0.048

The control removes the entire AGF module, including its residual projection, changing the encoder input width and parameter count. The ungated variant uses a fixed equal average of the graph and residual branches. The control and full-model rows use the same seed subset from Table 1. Errors are in K for temperature and percentage points for relative humidity.

Table 4. Gate-activation statistics pooled over all five CLCRN-AGF seeds and the full test set. The gate balances a graph-aggregated projection and a residual linear projection of the same input, before CLConv. Standard deviations describe the pooled population of scalar gates.

Variable	Mean μ	Std σ	Min	Max
Temperature	0.600	0.196	0.001	1.000
Relative humidity	0.570	0.244	0.001	1.000
Surface wind	0.508	0.130	0.049	0.954
Cloud cover	0.475	0.208	0.002	0.997

Each task includes $5 \times 657 \times 12 \times 2,048 \times 8 = 645,857,280$ scalar gates (seeds, test examples, input steps, nodes and channels). Channels are retained individually; no channel averaging or test-set subsampling is applied.